

Rakibul Islam

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Summary

I am a passionate AI and software engineer with hands-on experience in developing **intelligent systems**, **automation tools**, and **full-stack applications**. I have a proven track record of applying **machine learning (ML)**, **large language models (LLMs)**, and **deep learning** in real-world projects with publications in **IEEE** and **ACM**. Skilled in **rapid learning**, **problem-solving**, **deploying scalable systems**, and **leading teams**.

Skills

Artificial Intelligence & Machine Learning

- Multi-Agent Systems:** CrewAI, LangChain, LangGraph, LangSmith
- **Large Language Models (LLMs) & RAG:** Pre-training LLM, Fine-Tuning LLM, LlamaIndex, Hugging Face Pipeline, LLM APIs, Multimodal Retrieval-Augmented Generation
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, RNN, NLP, Sequence Modeling
- **Advanced AI Topics:** CNNs, GNNs, GANs, Autoencoders, AutoML, Transfer Learning
- **Computer Vision & Image Processing:** OpenCV, Vision Transformers, CNNs
- **AI Libraries & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Dash

Full Stack Development

- Backend Development:** ASP.NET Core 8 MVC (Repository, Unit of Work, Clean Architecture), ASP.NET Core Web API (Routing, Controllers, Views, Razor Pages, Dependency Injection), FastAPI, Django, RESTful API, API Security (JWT, OAuth2)
- **Frontend Development:** HTML, CSS, JavaScript, Bootstrap, Tailwind
- **Mobile Application Development:** Flutter
- **Web Scraping:** Selenium, BeautifulSoup, Scrapy, Playwright
- **Design Patterns & Architecture:** Creational, Structural, Behavioral, Concurrency, Architectural, Clean Architecture
- **Tools & Automation:** Docker, Apache Airflow (ETL), Apache Airflow (ETL), Linux CLI, Shell Scripting, Git, Azure
- **Data Structures & Algorithms**

Data Analysis & Databases

- Data Science:** Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis, Hypothesis Testing, A/B Testing, Time Series Analysis, Data Mining, Data Preprocessing
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Dash, Correlation Analysis, Trend Analysis, Reporting & Insights
- **Big Data & Storage:** Apache Spark, Apache Iceberg, AWS S3, Distributed Data Processing, Large-Scale Data Handling
- **ETL & Data Engineering:** ETL Pipelines, Apache Airflow, Data Pipelines, Automation, Data Validation, Data Integration
- **Databases & ORM:** SQL, PostgreSQL, Oracle, MongoDB, Firebase, SQL Optimization, ERD, Indexing, Stored Procedures, Entity Framework, SQLAlchemy, SQL Model
- **Productivity & Documentation:** CI/CD (GitHub Actions, Azure DevOps), PowerPoint, Excel, Technical Documentation

Experience

Software Programmer , Genuine Technology Research Limited (GTR) – Dhaka	Oct 2024 – Present
• Promoted quickly; developed features for Alfred and HRM software using ASP.NET MVC, integrating CrewAI, RAG, and LLMs to enhance automation.	
Python Intern , Genuine Technology Research Limited (GTR) – Dhaka	Aug 2024 – Oct 2024
• Enhanced system intelligence with ASP.NET MVC, automated workflows, developed APIs, and managed databases using modern tools like CrewAI, RAG, FastAPI, Docker, and Selenium.	
Co-Founder & CTO , Niharon Technologies – Dhaka	2022 – 2024
• Led technology strategy, product development, and team management for startup projects while collaborating with cross-functional teams and clients.	
General Member , UIU App Forum & UIU Entrepreneur Development Forum – United International University	2022 – 2024

Education

United International University , Dhaka — B.Sc. in Computer Science and Engineering — CGPA: 3.57 / 4.00	2020 – 2024
Gazipur Metropolitan College , Gazipur — Higher Secondary Certificate (HSC)	2019
Rani Bilashmoni Govt. Boys' High School , Gazipur — Secondary School Certificate (SSC) — GPA: 5.00/5.00	2017

Certificates

Machine Learning Specialization – Coursera, **IBM Data Science Professional Certificate**, **IBM AI Engineering Professional Certificate**, **Deep Learning Specialization** – Coursera, **Pretraining LLMs** – Advanced LLM Course, **Multi-AI Agent Systems with CrewAI**, **Statistics** – Google, **British Council IELTS English Certification**, **Presenting Author** – ICEEICT 2024 Conference

Publications

TIG-UNet++: A Thresholded Input Guided Modified UNet++ for Separating Lung Region in Chest X-Rays (ICTIS 2025, Bangkok)	2025
Deep Multi-Modal Approach for Protein Function Prediction and Classification (IEEE)	2024
From Pixels to Diagnosis: Convolutional Neural Networks in Tuberculosis Screening (IEEE)	2024
Risk Minimization by Early Readmission Prediction for Diabetes Patients: A Data-Driven Machine Learning Approach (IEEE)	2024
Ensemble Deep Learning for Real-Time Textile Fabric Defect Detection and Classification Using YOLOv8 and CNNs (ACM)	2024
Co-relation-based Feature Extraction to Improve Classification Accuracy (IEEE)	2024
Smart Grid Stability Prediction for Efficient Power Management Using Machine Learning (IEEE)	2024

Projects

Fabric Defect Detection and Classification

- Used ensemble of CNNs and YOLOv8 for real-time defect detection in textile images
- Tools Used: Python, OpenCV, TensorFlow, CNN, YOLOv8

Job Application Automation System

- Automated job application using LLMs, RAG, CrewAI, and FastAPI backend
- Tools Used: .NET MVC 8, Ollama, FastAPI, CrewAI, Langchain

HRM Software & Project Management Software

- Developed employee management and projects management system with dashboarding and CRUD
- Tools Used: .NET MVC 8, Bootstrap, JS, Tabulator

Other Key Projects

- **Weapons Object Detection, Tuberculosis Screening, Face Recognition, Diabetic Retinopathy Detection** — CNN, Transfer Learning
- **Protein Function Prediction, Sentiment Analysis, Bengali Text to IPA** — NLP, DNN, Visualization
- **Customer & Stock Analysis, House Pricing Analysis** — ML, EDA, Dashboarding
- **House Rental Website, Auto Code Evaluator** — Django, SQL, Web Dev
- **Domain-Specific AI Assistants** — RAG, Langchain, LlamaIndex

Technologies

Languages: Python, C, C++, C#, JavaScript, SQL, HTML, CSS

Frameworks/Tools: ASP.NET MVC/Core 8, Django, FastAPI, Flask, Tabulator, Bootstrap, OpenCV, TensorFlow, Keras, PyTorch, Numpy, Pandas, Scikit-learn, SQLAlchemy

Technologies: Langchain, CrewAI, RAG, Ollama, Hugging Face, LlamaIndex, Docker, Apache Airflow, Selenium, Git